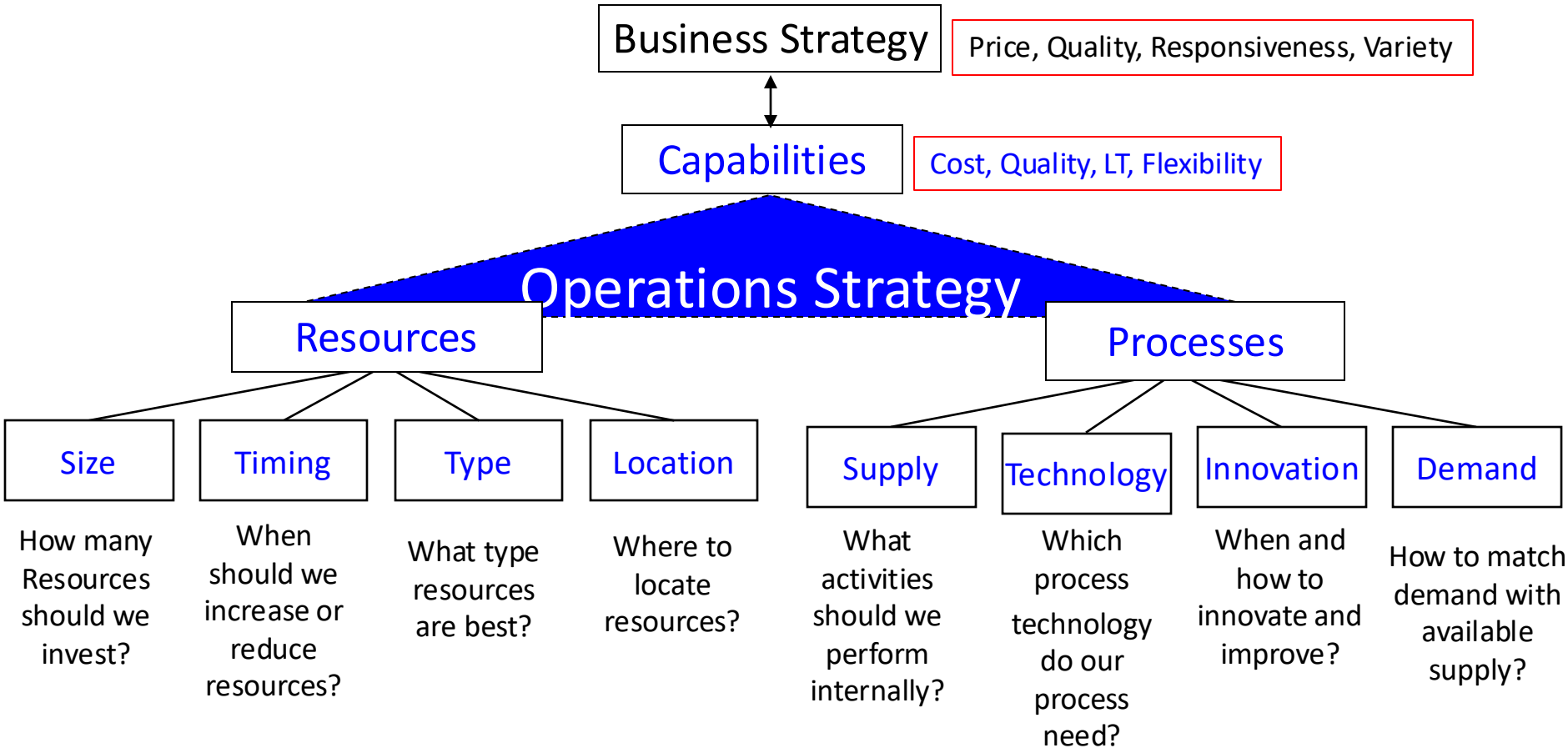
# Today's Learning Objectives

Be able to

- Explain the concept of operational trade-offs and discuss its impact on competition
- Analyze competitive threats using trade-off curves
  - Introduce trade-off curves, Efficient Frontier, Operational Efficiency (OE)
- Explain what operational focus is and how it improves operational efficiency (OE)
- Design operations strategies to improve competitive advantage (e.g., tailor the right degree of operational focus)

# OS Framework






(Source: Operations Strategy by Prof. Jan A. Van Mieghem)

# Concept of operational Trade-offs

- Trade-off is "a balancing of factors all of which are not attainable at the same time; a giving up of one thing in return for another" - Merriam-Webster (2007)
- Operational trade-offs: operational capabilities (i.e., Cost, Quality, LT, Flexibility) are governed by trade-offs that are defined by the resources and the processes, i.e., operational systems
  - It says that an organization can't have it all because its capabilities exhibit trade-offs due to the finite abilities of its operational system
  - Alternatively speaking, operations strategy can shape and change the trade-offs by choosing the <u>right</u> operational systems
  - Hence, operational trade-offs require an organization to make choices

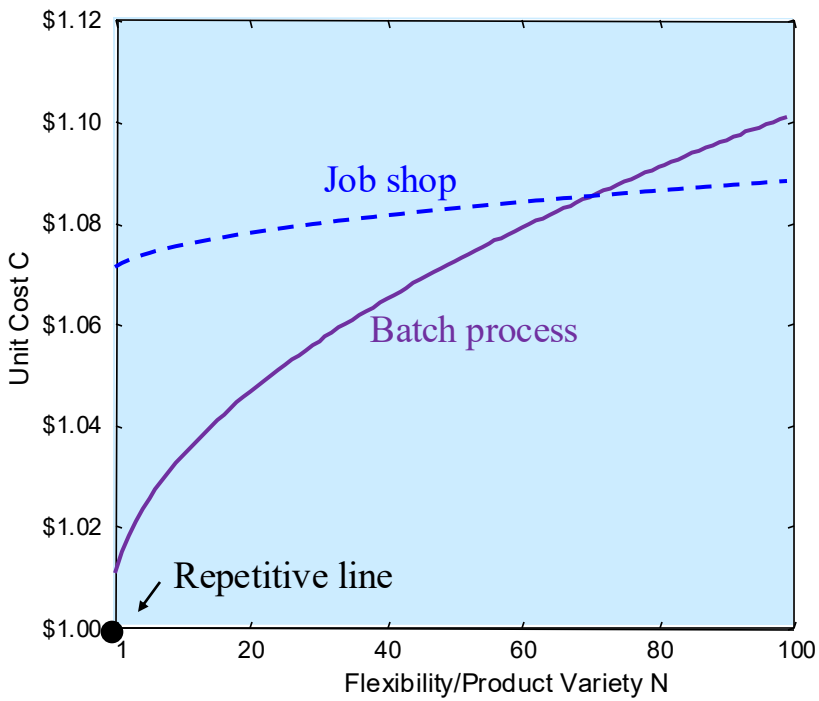
# Operational systems: ACC vs DJC

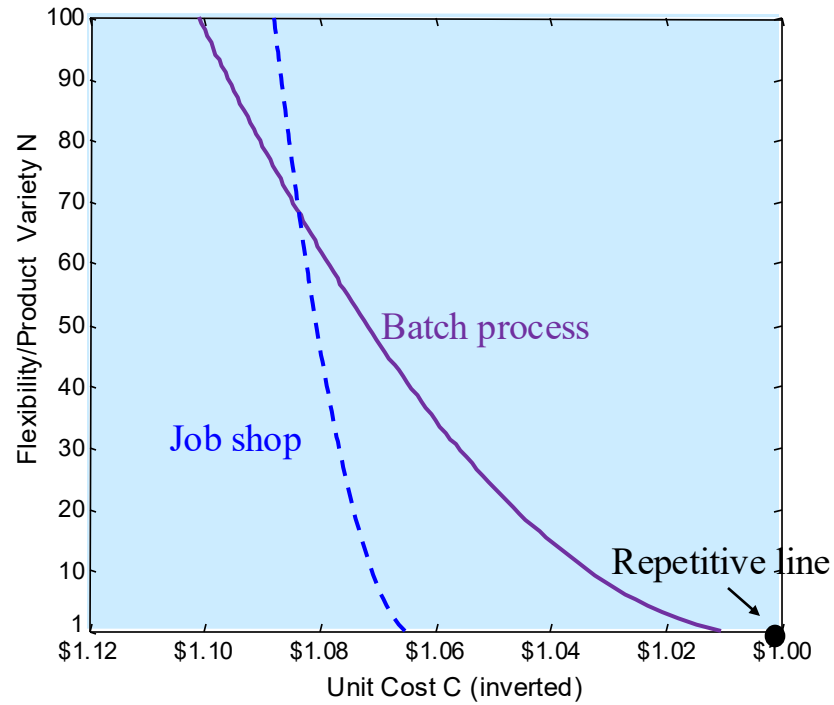
# Trade-off curves example

- The choice of operational system shapes a company's capability (i.e., Cost, Quality, LT, Flexibility) trade-offs
- A trade-off curve is the graphical representation of the trade-offs
- Operational trade-offs can be presented by iso-NPV curves
  - These curves (or spaces, or …) would show the capability combinations (i.e., Cost, Quality, LT, Flexibility) that yield a given NPV
  - Or call it indifference curves for the same NPV
- For simplicity, we will only consider 2-dimensional trade-off curves

# Trade-off curves example

- The choice of operational system shapes a company's capability trade-offs
  - Job shop: highest unit cost, lowest changeover cost
  - Batch: in between
  - Repetitive line: lowest unit cost, highest changeover costs

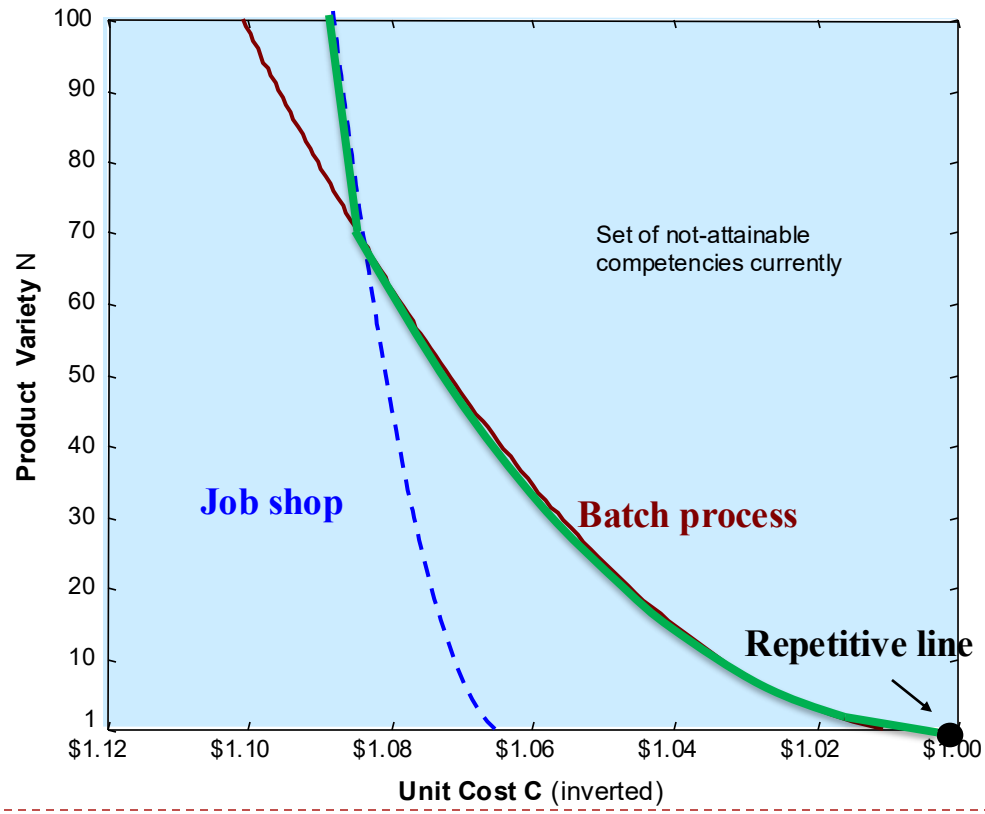


# Efficient Frontier

- The efficient frontier is a snapshot that traces the industry's lowest costs to produce specific non-cost-based capability bundles, e.g., quality, LT, and flexibility

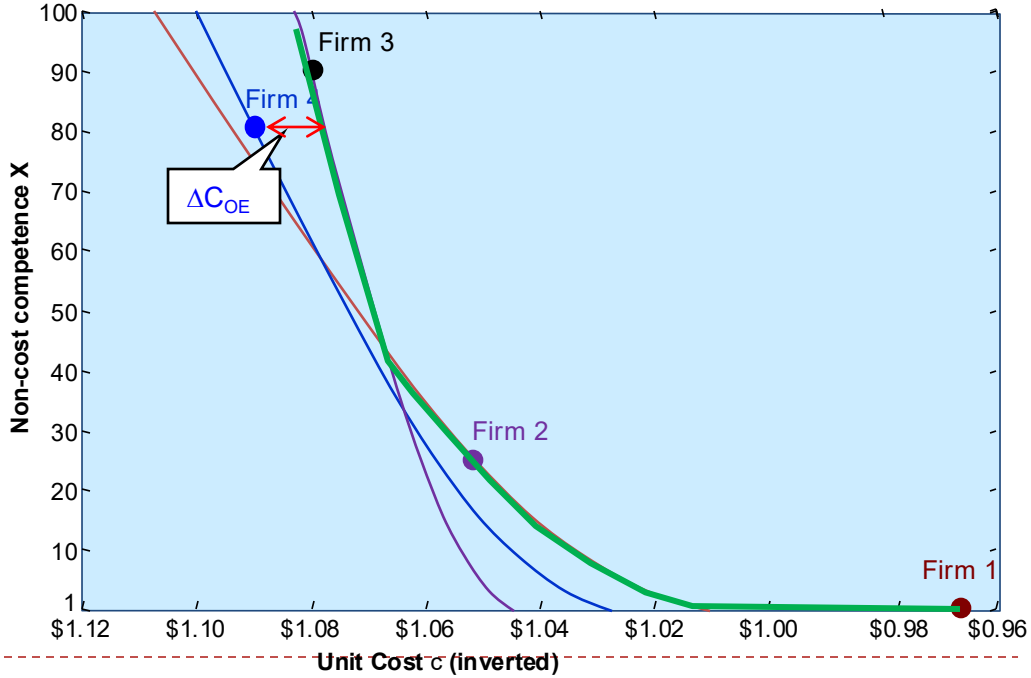
- Each operational system has a "sweet spot", an area on its trade-off curve where it operates on the efficient frontier



# Operational Efficiency (OE)

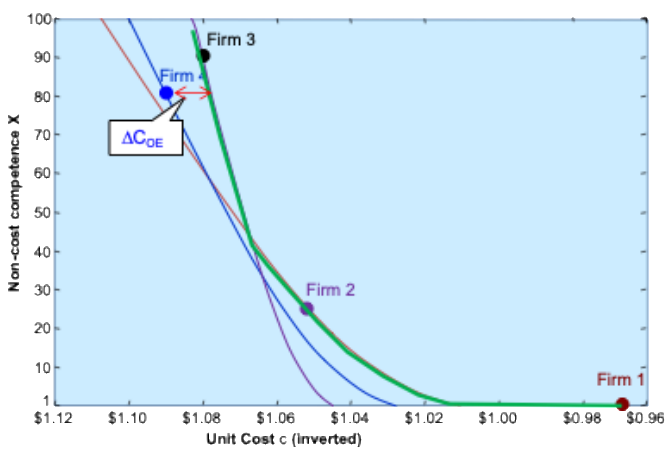
- A non-cost-based capability's operational efficiency (OE) is measured by its cost disadvantage relative to the efficient frontier for that same capability
  - Implications: we can use this concept to analyze threats by estimating the largest cost advantage ($\Delta C_{OE}$) any of our competitors would have over us were they to use their operational systems to deliver the same (non-cost) capabilities as we do

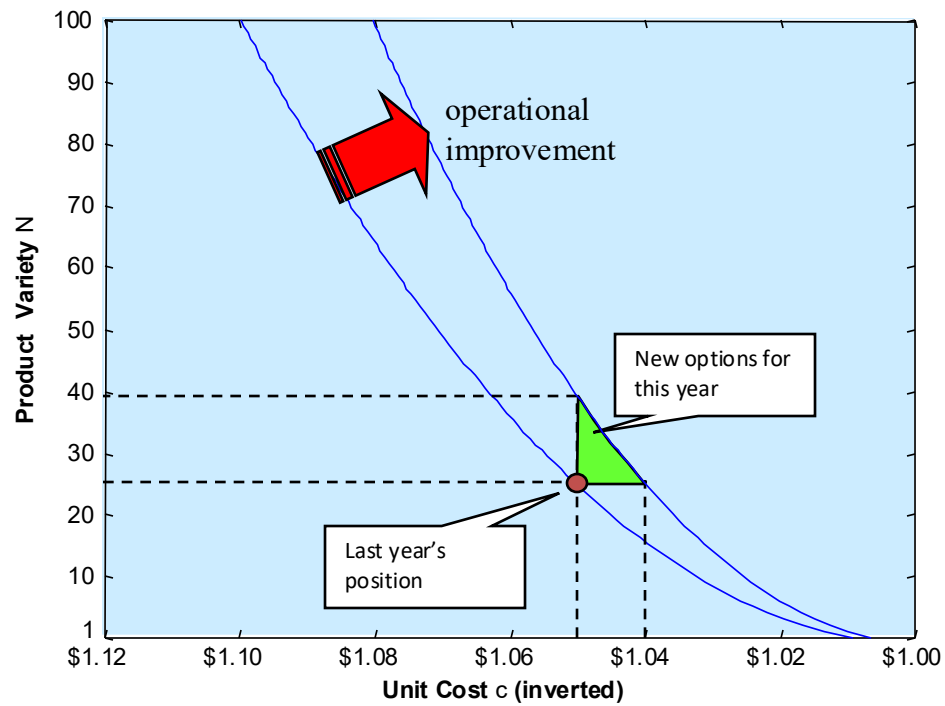
1. If this cost advantage of our most threatening competitor is zero or negative, we are operationally efficient
2. Otherwise, we are operationally inefficient at the rate of the cost disadvantage of our most threatening competitor



# OE improvement

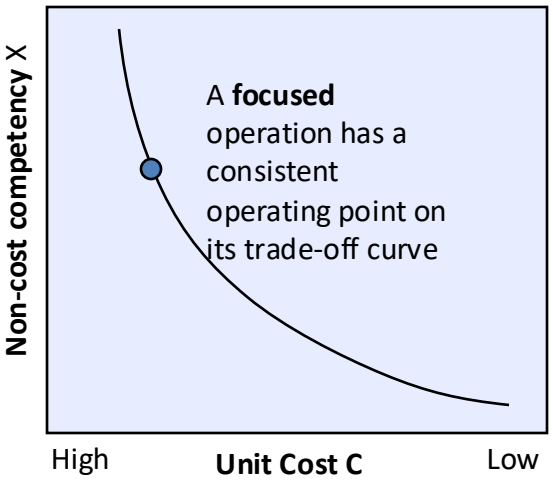
- OE improvement moves trade-off curve outwards
  - e.g., setup time reduction creates options to either increase differentiation or cost-efficiency (below right)
    - both can lead to reduction of $\Delta C_{OE,}$ if positive to start with

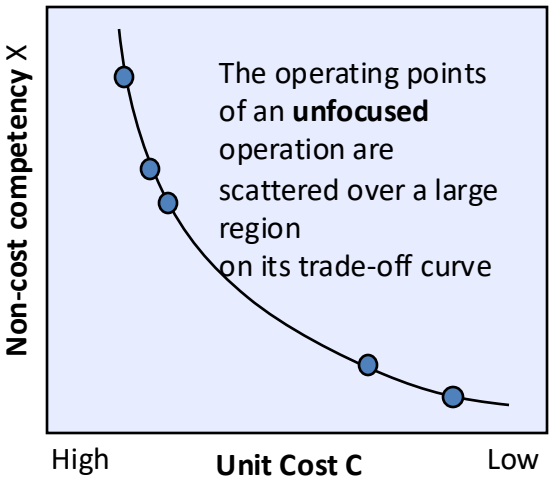


# How to improve OE?

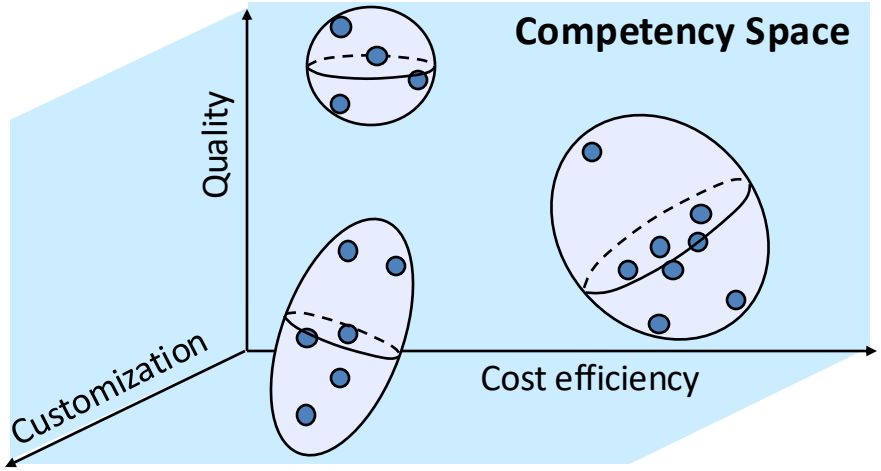
- Operational **focus** increases operational efficiency (OE).
  - A focused operation is one whose required capabilities are restricted to a narrow set in the capability space of costs, quality, LT, and flexibility.
  - Simplicity and repetition breeds competence:
    - learning and optimization
    - less divergence in attention spans
- Operational focus can be phrased in terms of points on the trade-off curve

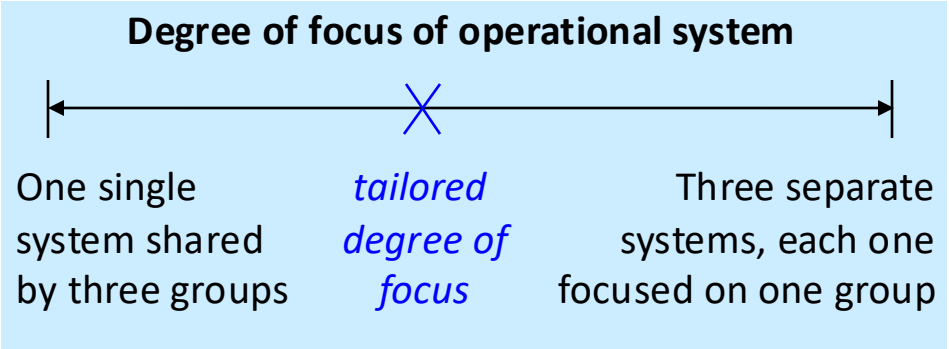


# How to **focus**? divide and conquer!

1. Partition products and services into groups with homogeneous capability needs (The example to the right has 3 groups).
2. For each group, separate those resources and processes that critically support its key value proposition while sharing the others
3. Create a charter for each operation to retain focus <u>over time</u>



# Who is more focused? ACC or DJC?

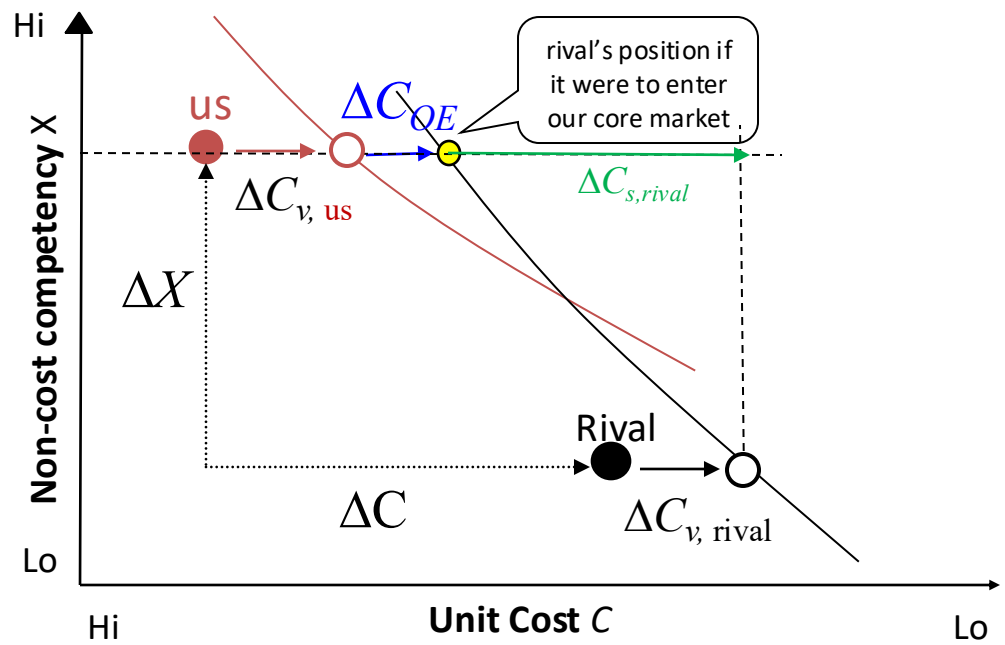
# Challenges related to operational focus

- A pressure to grow will lead to the inclusion of products and services with different customer needs.
  - Example: Dell selling laptop computers
- Changes in customer needs or competitors' positions may lead to the use of operational systems in ways they were not designed for.
  - Example: Marks & Spencer trying to deliver fast fashion
- Operational focus does not require a narrow product mix as is often assumed. It is quite possible to focus on variety and many firms successfully do so
  - Example: three-star restaurants (Michelin Guide) have been constantly changing menus but are very focused in what they try to achieve
- Operational efficiency is necessary but not sufficient for NPV maximization, it should also
  - Be sufficiently different from competitors to avoid direct competition
  - Be difficult to copy by competitors
  - Produce products/services that are preferred by a large enough customer segment (e.g., SWISS watcher makers in the 70s)
  - If it can leverage large economies of scale, one combined unfocused system may still generate more value than two separated focused systems.
    - An essential question is to what extent assets and processes can be shared?
    - What are benefits and costs of sharing? Economy of scale and scope?
- Operational focus can lead to rigidity
  - Example: Ford had a hard time adjusting to changes in customer trade-offs after its highly focused Model T success (in 1908)

# Who is more operational efficient? ACC or DJC?

# Analyzing threats using Trade-off curves

- Differentiation strategy vs. Low-cost position
- Adjusting economic conditions yields costs difference due to volume
- Adjusting operating conditions with given processes and resources yield trade-off curves
- Breaking up ΔC into volume, strategy and OE components, we have
  - ✓ $\Delta C + \Delta C_{v,\,\text{rival}} = \Delta C_{v,\text{us}} + \Delta C_{\text{OE}} + \Delta C_{\text{S}}$
  - ✓ $\Delta C_{\text{OE}} = \Delta C + \Delta C_{v,\,\text{rival}} - \Delta C_{v,\text{us}} - \Delta C_{\text{S}}$



# Analyzing threats using Trade-off curves

- How to estimate $\Delta C_{OE}$?
  - $\Delta C_{OE}$ is the operational efficiency-driven cost differential
    - It measures how one company is just more efficient in their operations systems than another, even if both were to run at their desired utilizations and to provide identical value proposition
    - To estimate $\Delta C_{OE}$, we must control for location, volume, and strategy
    - the seriousness of the threat is directly related to $\Delta C_{OE}$
  - $\Delta C_v$ is the volume-(or utilization) cost differential
    - Utilization-driven cost differential refers to the extra costs incurred by operating at less than the strategically-targeted utilization level (e.g. economy down, market miss/decline) because of spreading a fixed cost over fewer units
    - We need to consider both ours and our rivial's ($\Delta C_{v,us}$ and $\Delta C_{v,\ rival}$)
  - $\Delta C_S$ is the competitive strategy-driven cost differential
    - This is the cost difference inherent in the way each company chooses to compete
    - We would expect that a strategy emphasizing product innovation and customer responsiveness is more costly than a pure low-cost strategy (e.g. customized products would cost more than standardized ones).

# Case: Analyzing threats using Trade-off curves

Overall cost differences static comparison in 1991 (refer to Exhibits 7 & 8)

| ITEM | DJC@ Japan | INDEX | DJC @US | ACC |
|---|---|---|---|---|
| RM Product | $12.13 | 0.6 | $7.28 | $9.39 |
| RM Packaging | 2.76 | 0.6 | 1.66 | 2.10 |
| Labor (total) | 3.77 | 1.1 | 4.15 | 10.30 |
| Electricity | 1.40 | 0.8 | 1.12 | 0.80 |
| Depreciation | 1.80 | 1.0 | 1.80 | 5.10 |
| Other | 4.24 | 1.0 | 4.24 | 6.10 |
| Total | $26.10 | | $20.25 | $33.79 |

$\Delta$ C = $7.69

$\Delta$ C = $13.54

Major DJC Savings of the $\Delta$C=$13.54 are in

- RM: product and packaging design
- Labor: automation and multi-functional training
- Depreciation: high utilization of capacity
- Other: case did not mention what this is, perhaps overhead?

Question: how much of the $\Delta$C=$13.54 is due to

- volume being different from targeted volumes
  - ACC's real capacity utilization is 70% (while planned at 85%) $\Delta C_{v,ACC}$
  - DJC's real capacity utilization is 87.5% (while planned at 100%) $\Delta C_{v,\,DJC}$
- strategic positioning $\Delta C_S$
- operational (in)efficiency (OE) in RM, Labor, Depreciation and other

# Case: Analyzing threats using Trade-off curves

- Costs differences in Labor: $\Delta C_{Labor}$ = 10.30 – 4.15 = \$6.15 (refer to previous slide)
  - One approach to estimate volume effects is to look at the cost of labor per unit:
    - Total ACC labor bill is \$10.30/thousand x 420 million units (70% utilization) = \$4.3M.
      - If ACC would operate at 85% capacity utilization (designed strategy) with same amount of labor, labor costs would be \$4.3/510 million = \$8.43/ thousand, thus
      - $\Delta C_{labor, v, ACC}$ = \$10.30 –\$8.43 = \$1.87
    - Total DJC labor bill is \$4.15/thousand x 700 million units (87.5% utilization) = \$2.905M.
      - If DJC would operate at 100% capacity utilization (designed strategy) with same amount of labor, labor costs would be \$2.905/800million = \$3.63/thousand, thus
      - $\Delta C_{labor, v, DJC}$ = \$4.15 - \$3.63 = \$0.52
  - Use same cost of labor per unit approach to estimate strategy effects:
    - If DJC enters US market and would operate at 85% capacity utilization like ACC with same amount of labor, labor costs would be \$2.905/(0.85*800) million = \$4.27/ thousand, thus
      - $\Delta C_{labor, s, DJC}$ = \$4.27 - \$3.63 = \$0.64
  - Thus, operations efficiency cost differential
    - $\Delta C_{OE} = \Delta C_{Labor} + \Delta c_{labor,v,DJC} - \Delta c_{labor,v,ACC} - \Delta c_{labor,s,DJC}$ = 6.15 + 0.52 – 1.87 – 0.64 = \$4.16

# Guidelines for Operations-based Competition

- Study your own as well as your rivals' trade-off curves
  - So, we can do threat analysis such as estimating $\Delta C_{OE}$
  - Be careful, we need to adjust for location, volume or utilization, and strategy before concluding $\Delta C_{OE}$
- Strength your competitive position
  - Focus, manage suppliers' and buyers' relations, labour and capital markets … such as to increase $\Delta C_S$ and thereby reduce $\Delta C_{OE}$
- Anticipate customer dynamics
  - A longer-term strategy should include market research to estimate customer utility dynamics
  - This is to manage $\Delta C_v$

# Summary of Learning Objectives

- Operational trade-off
  - Simple adjustments in the use of operational systems will reveal trade-offs between the performance on different competency dimensions
  - Across existing operational systems in an industry trade-offs between competencies exist (efficient frontier)
  - Operational systems provide competitive advantage when their use is operationally efficient and generates products that large enough customers segment are willing to pay for; and when they are sufficiently different from and hard to copy by rivals
- Analyze a competitive threat using trade-off curves
  - Don't use cost numbers at face value. Rather, adjust for location, strategic positioning and volume (utilization) differences to arrive at the cost advantage due to operational efficiency, which is the meaningful "net cost advantage" number.
- Operational focus
  - Is achieved when an operational system is used to deliver a narrow set of competency bundles (does not require a narrow product mix)
  - Leads to operational efficiency because practice perfects and no compromises are necessary in the design of the operational system
  - Is hard to maintain (pressure to grow, customer needs change over time,
  - Is no guarantee for success (focus is necessary but not sufficient)
  - can be applied judiciously to maximize NPV (Ford Model T)
  - one combined unfocused system may generate more value than two separated focused systems
- Design strategies to improve competitive advantage through operations:
  - Use focus judiciously, adopt best practices from competitors and develop proprietary practices in support of a differentiated and attractive value proposition